HUNNY TUTORIALS

YOUR CBSE PARTNER

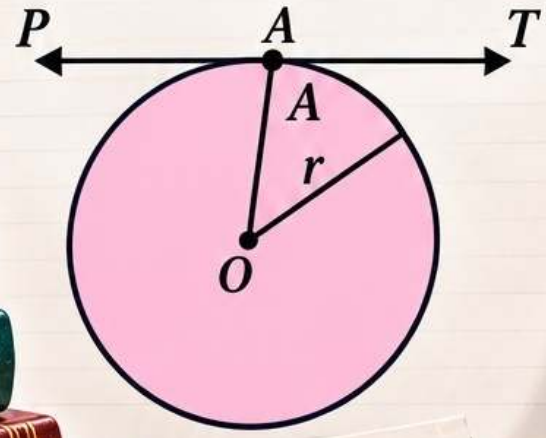
Class X Mathematics

2026 - 27

Unit - Geometry

Weightage in Boards Exam - 15 Marks

Ch - 10 Circles



- Tangent to a Circle
- Number of Tangents
- Length of Tangents
- Concyclic Points

CIRCLES

GEOMETRY

MATHEMATICS

Free NCERT Solutions

Just revise it then solve it.

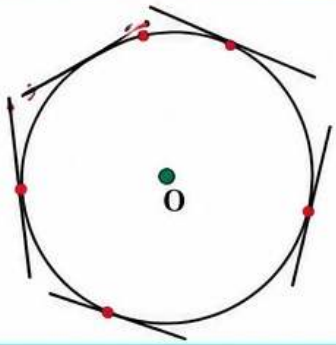
Prepared by **Hunny Sir** | 9999903256

Exercise 10.1

Circles

Q1. How many tangents can a circle have?

Sol. A circle can have an **infinite** number of tangents.



Concept Clarity

- A tangent is a straight line that touches a circle at **exactly one point**.
- Since a circle has infinitely many points on its circumference, there can be infinitely many tangents.

Q2. Fill in the blanks.

(i) A tangent to a circle intersects it in ..... point(s).
(ii) A line intersecting a circle in two points is called .....
(iii) A circle can have ..... parallel tangents at the most.
(iv) The common point of a tangent to a circle and the circle is called .....

Sol.

(i) Exactly one.
(ii) Secant.
(iii) Two.
(iv) Point of contact.

Concept Clarity

- A tangent meets a circle at exactly one point.
- A secant is a line that intersects a circle at two points.
- A circle can have **two parallel tangents** at the most (on opposite sides of the circle).
- The common point of the tangent and the circle is called the **point of contact**.

Q3. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that OQ = 12 cm. Length of PQ is:

(a) 12 cm **(b) 13 cm** **(c) 8.5 cm** **(d) $\sqrt{119}$ cm**

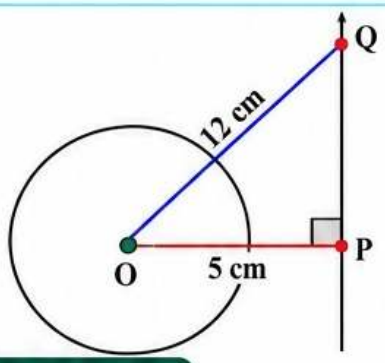
Sol.

Given, radius of the circle, OP = 5 cm.
OQ = 12 cm.
Since the **radius** at the **point** of contact is perpendicular to the tangent,
$\angle OPQ = 90°$.
In right $\triangle OPQ$, by Pythagoras theorem:

$$PQ^2 = OQ^2 - OP^2$$
$$= 12^2 - 5^2$$
$$= 144 - 25$$
$$= 119$$

$\therefore$ $PQ = \sqrt{119}$ cm.

Hence, the correct option is (d).



Concept Clarity

- The radius drawn to the point of contact is perpendicular to the tangent.
- Hence, $\triangle OPQ$ is a right-angled triangle and we use Pythagoras theorem to find PQ.

Exercise 10.2

Circles

In Q.1 to 3, choose the correct option and give justification.

Q1. **From a point Q, the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. The radius of the circle is**

(A) 7 cm **(B)** 12 cm **(C)** 15 cm **(D)** 24.5 cm

Sol.

Let O be the centre of the circle, P the point of contact and Q the external point.

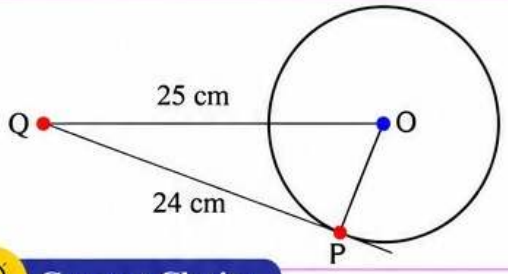
Given: QP = 24 cm, OQ = 25 cm.

In right ΔOPQ, $(\angle OPQ = 90°)$

$$OQ^2 = OP^2 + PQ^2$$
$$(25)^2 = OP^2 + (24)^2$$
$$625 = OP^2 + 576$$
$$OP^2 = 49$$
$$OP = 7 \text{ cm}$$

Therefore, the radius of the circle is 7 cm.

Hence, the correct option is **(A)**.



Concept Clarity

- The length of the tangent from an external point to a circle can be found using Pythagoras theorem: $(\text{Distance from centre})^2 = (\text{Radius})^2 + (\text{Tangent})^2$.
- The radius at the point of contact is perpendicular to the tangent.

Q2. **In the figure, if TP and TQ are the two tangents to a circle with centre O so that $\angle POQ = 110°$, then $\angle PTQ$ is equal to**

(A) 60° **(B)** 70° **(C)** 80° **(D)** 90°

Sol.

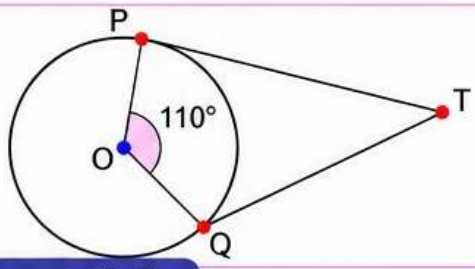
TP and TQ are tangents, so $OP \perp TP$ and $OQ \perp TQ$.

Thus, $\angle OPT = 90°$ and $\angle OQT = 90°$.

In quadrilateral OPTQ,

$$\angle PTQ + \angle POQ + \angle OPT + \angle OQT = 360°$$
$$\angle PTQ + 110° + 90° + 90° = 360°$$
$$\angle PTQ + 290° = 360°$$
$$\angle PTQ = 70°$$

Hence, the correct option is **(B)**.



Concept Clarity

- The angle between two tangents from an external point is supplementary to the angle subtended by the points of contact at the centre.
- $\angle PTQ + \angle POQ = 180°$.

Q3. **If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80°, then $\angle POA$ is equal to**

(A) 50° **(B)** 60° **(C)** 70° **(D)** 80°

Sol.

Tangents PA and PB are inclined at 80°, so $\angle APB = 80°$.

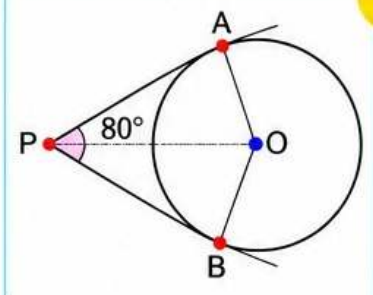
In quadrilateral AOBP,

$$\angle APB + \angle AOB = 180°$$
$$80° + \angle AOB = 180°$$
$$\angle AOB = 100°$$

ΔOAP and ΔOBP are congruent (RHS), so

$$\angle POA = \angle POB = \frac{1}{2}\angle AOB$$
$$\angle POA = \frac{1}{2} \times 100° = 50°$$

Hence, the correct option is (A).



Concept Clarity

- The angle between two tangents from an external point is **supplementary** to the central angle between the points of contact.
- $\angle APB + \angle AOB = 180°$ and $\angle POA = \frac{1}{2}\angle AOB$.

Exercise 10.2

Q4. **Prove that the tangents drawn at the ends of a diameter of a circle are parallel.**

Sol.

Let AB be a diameter of the circle with centre O.
Let p and q be the tangents at A and B respectively.

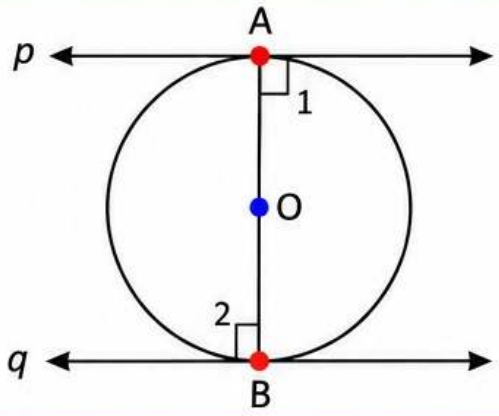
Since a radius at the point of contact is perpendicular to the tangent,

$OA \perp p \Rightarrow \angle 1 = 90°$
$OB \perp q \Rightarrow \angle 2 = 90°$

Here, $\angle 1$ and $\angle 2$ are alternate angles and $\angle 1 = \angle 2 = 90°$.

Therefore, $p \parallel q$ (If a pair of alternate angles are equal, then the lines are parallel).

Hence, the tangents drawn at the ends of a diameter of a circle are parallel.



Concept Clarity

- The tangent at any point of a circle is perpendicular to the radius through the point of contact.
- If a pair of alternate angles are equal, then the two lines are parallel.

Q5. **Prove that the perpendicular at the point of contact to the tangent to a circle passes through the centre.**

Sol.

Let XY be a tangent to the circle with centre O and radius r, touching the circle at B.
Let AB be a line perpendicular to XY at B.

Join OB.

Since the radius at the point of contact is perpendicular to the tangent,

$\angle OBX = 90°$ and $\angle OBY = 90°$
[X, B, Y are collinear]

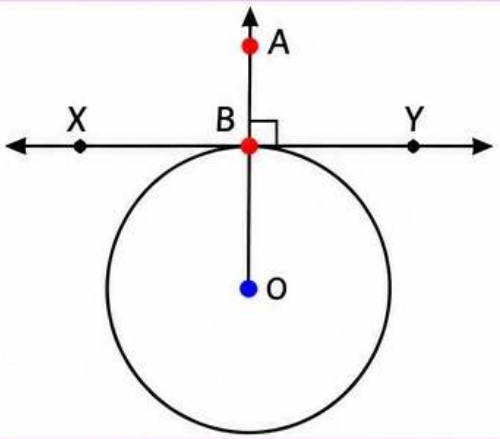
Thus, $\angle ABY = 90°$

Now, $\angle ABY + \angle OBY = 180°$ [Linear pair]

$\Rightarrow 90° + 90° = 180°$

$\Rightarrow$ A, B, O are collinear.

Hence, the perpendicular at the point of contact to the tangent to a circle passes through the centre.



Concept Clarity

- The radius at the point of contact is perpendicular to the tangent.
- If a line is perpendicular to a tangent at the point of contact, then it passes through the centre of the circle.

Exercise 10.2

Q6. **The length of a tangent from a point A at a distance 5 cm from the centre of the circle is 4 cm. Find the radius of the circle.**

Sol.

Let O be the centre of the circle and P be the point of contact of the tangent from A.

$OA = 5$ cm (distance from A to centre)

AP = 4 cm (length of tangent)

Let OP = r cm be the radius of the circle.

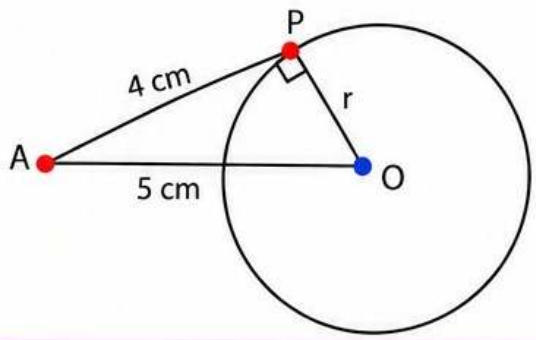
Since radius at the point of contact is perpendicular to the tangent,

$\angle OPA = 90^\circ$

In right ΔOPA, by Pythagoras theorem:

$$OA^2 = AP^2 + OP^2$$
$$5^2 = 4^2 + r^2$$
$$25 = 16 + r^2$$
$$r^2 = 25 - 16 = 9$$
$$r = \sqrt{9} = 3 \text{ cm}$$

Hence, the radius of the circle is 3 cm.



Concept Clarity

- A tangent to a circle is perpendicular to the radius through the point of contact.
- In the right triangle formed, use Pythagoras theorem to relate the distance from the external point, the tangent length and the radius.

Q7. **Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.**

Sol.

Let O be the common centre of the two concentric circles of radii 5 cm (larger circle) and 3 cm (smaller circle).

Let AB be a chord of the larger circle which touches the smaller circle at P.

OP = 3 cm (radius of smaller circle)

OA = 5 cm (radius of larger circle)

Since the radius at the point of contact is perpendicular to the chord,

OP $\perp$ AB

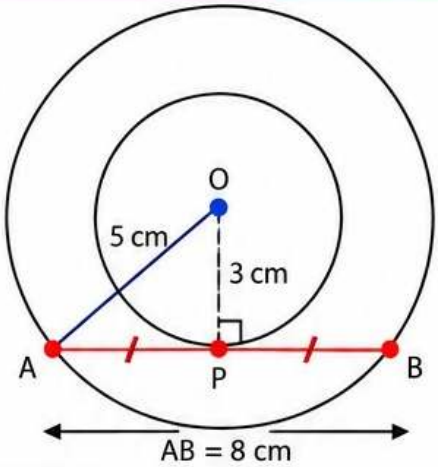
Therefore, OP bisects AB, so AP = PB.

In right ΔOAP, by Pythagoras theorem:

$$OA^2 = OP^2 + AP^2$$
$$5^2 = 3^2 + AP^2$$
$$25 = 9 + AP^2$$
$$AP^2 = 16 \Rightarrow AP = 4 \text{ cm}$$

So, AB = 2 × AP = 2 × 4 = 8 cm

Hence, the length of the chord is 8 cm.



Concept Clarity

- A chord which touches a concentric circle is perpendicular to the radius at the point of contact and is bisected by it.
- Use Pythagoras theorem in the right triangle formed to find half the chord, then double it to get the chord length.

Exercise 10.2

Q8. **A quadrilateral ABCD is drawn to circumscribe a circle (see figure). Prove that: AB + CD = AD + BC**

Sol.

Let the points of contact of the incircle with AB, BC, CD and DA be P, Q, R and S respectively.

From the property that tangents drawn from an external point to a circle are equal:

AP = AS ...(i)

BP = BQ ...(ii)

CR = CQ ...(iii)

DR = DS ...(iv)

Adding (i) to (iv), we get

AP + BP + CR + DR = AS + BQ + CQ + DS

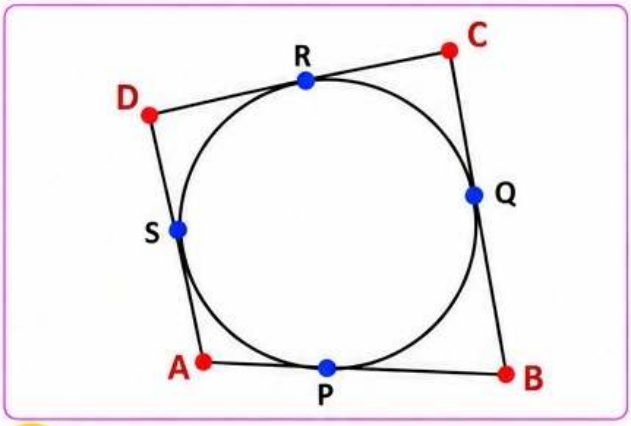
Rearranging,

(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ)

That is, AB + CD = AD + BC

Thus, AB + CD = AD + BC, as required.

Hence, Proved.



Concept Clarity

- Tangents drawn from an external point to a circle are equal in length.
- In a quadrilateral that circumscribes a circle (tangential quadrilateral), the sums of the lengths of opposite sides are equal.

Q9. **In the figure, XY and X′Y′ are two parallel tangents to a circle with centre O and another tangent AB with point of contact C intersecting XY at A and X′Y′ at B. Prove that ∠AOB = 90°.**

Sol.

Given: XY || X′Y′ are two parallel tangents to a circle with centre O. Tangent AB with point of contact C intersects XY at A and X′Y′ at B.

To Prove: ∠AOB = 90°.

Construction: Join OA, OB and OC.

Proof: In ΔAOP and ΔAOC

AP = AC [Lengths of tangents from an external point]

OP = OC [Radii of the circle]

OA = OA [Common]

∴ ΔAOP ≅ ΔAOC [SSS congruence rule]

⇒ ∠PAO = ∠CAO [C.P.C.T.]

⇒ ∠PAC = 2∠OAC ...(i)

Similarly, ∠QBC = 2∠OBC ...(ii)

Since XY || X′Y′, ∠PAC + ∠QBC = 180°

[Interior consecutive angles on the same side of transversal]

⇒ 2(∠OAC + ∠OBC) = 180°

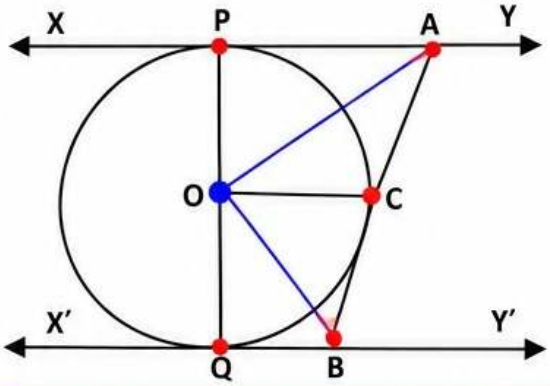
⇒ ∠OAC + ∠OBC = 90°

In ΔAOB, ∠AOB + (∠OAC + ∠OBC) = 180°

⇒ ∠AOB + 90° = 180°

⇒ ∠AOB = 90°

Hence, ∠AOB = 90°.



Concept Clarity

- The radius at the point of contact is perpendicular to the tangent.
- Tangents drawn from an external point are equal.
- Use properties of parallel lines and interior consecutive angles.
- Congruent triangles help relate the angles formed with the centre.

Exercise 10.2

Q10. **Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line-segment joining the points of contact at the centre.**

Sol.

Let PA and PB be two tangents to the circle with centre O and A, B be the points of contact.

$OA \perp AP$ and $OB \perp BP$

$\angle OAP = \angle OBP = 90°$

[Radius and tangent are perpendicular to each other]

In quadrilateral OAPB,

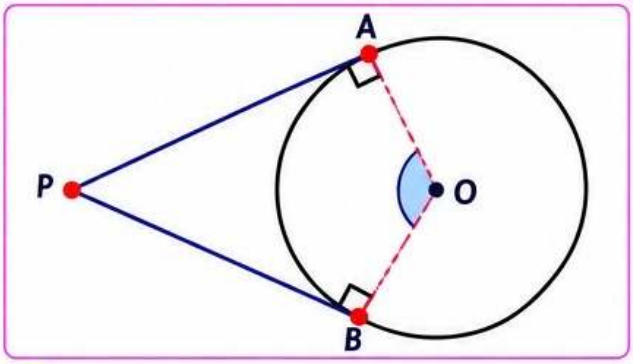
$\angle OAP + \angle OBP + \angle APB + \angle AOB = 360°$

$\Rightarrow 90° + 90° + \angle APB + \angle AOB = 360°$

$\Rightarrow 180° + \angle APB + \angle AOB = 360°$

$\angle APB + \angle AOB = 360° - 180° = 180°$

Hence, the angle between the two tangents ($\angle APB$) is supplementary to the angle subtended by AB at the centre ($\angle AOB$).



Concept Clarity

- The radius at the point of contact is perpendicular to the tangent.
- In quadrilateral formed by centre, points of contact and external point, use angle sum property (360°).
- The angle between two tangents from an external point is supplementary to the central angle subtended by the line joining the points of contact.

Q11. **Prove that the parallelogram circumscribing a circle is a rhombus.**

Sol.

Let ABCD be a parallelogram circumscribing a circle with centre O, touching AB, BC, CD and DA at P, Q, R and S respectively.

$OP \perp AB$ and $OS \perp AD$

In ΔOPB and ΔOSD,

$\angle OPB = \angle OSD = 90°$

$OB = OD$ [Diagonals of a parallelogram bisect each other]

$OP = OS$ [Radii]

$\therefore \Delta OPB \cong \Delta OSD$ [RHS congruence rule]

$\Rightarrow PB = SD$ [C.P.C.T.] ...(i)

Also, $AP = AS$ [Lengths of tangents from an external point] ...(iii)

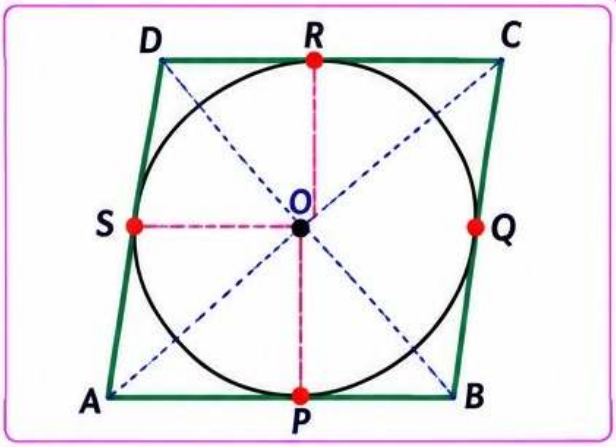
Adding (i) and (ii),

$AP + PB = AS + SD$

$\Rightarrow AB = AD$

Similarly, we can show $AB = BC = CD = DA$.

Hence, the parallelogram ABCD is a rhombus.



Concept Clarity

- In a parallelogram, diagonals bisect each other.
- The perpendicular from the centre to each side meets the side at the point of contact (radius $\perp$ tangent).
- Lengths of tangents from an external point are equal.
- Using RHS congruence, we get opposite sides equal, and hence all four sides are equal, so the parallelogram is a rhombus.

Exercise 10.2

Circles

Q12. **A triangle ABC is drawn to circumscribe a circle of radius 4 cm such that the segments BD and DC into which BC is divided by the point of contact D are of lengths 8 cm and 6 cm respectively (see figure). Find the sides AB and AC.**

Sol.

Let the circle touch AB, BC and CA at E, D and F respectively.

Given : BD = 8 cm, DC = 6 cm, $r = 4$ cm

By the property of equal tangents from an external point:

BE = BD = 8 cm, CD = CF = 6 cm

Let AE = AF = x cm

Then, AB = AE + EB = $x + 8$ and AC = AF + FC = $x + 6$

Also, BC = BD + DC = 8 + 6 = 14 cm

Semi-perimeter, $s = \frac{a+b+c}{2} = \frac{14+(x+8)+(x+6)}{2}$

$= \frac{2x+28}{2} = (x+14)$ cm

Area of ΔABC (by Heron's formula)

$= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{(x+14)(x)(8)(6)} = \sqrt{48x(x+14)}$ ...(i)

Since ΔABC is formed by three triangles with the centre O,

Area of ΔABC = ar(ΔOBC) + ar(ΔOCA) + ar(ΔOAB)

$= \frac{1}{2} \times 4 \times 8 + \frac{1}{2} \times 4 \times 6 + \frac{1}{2} \times 4 \times 14$

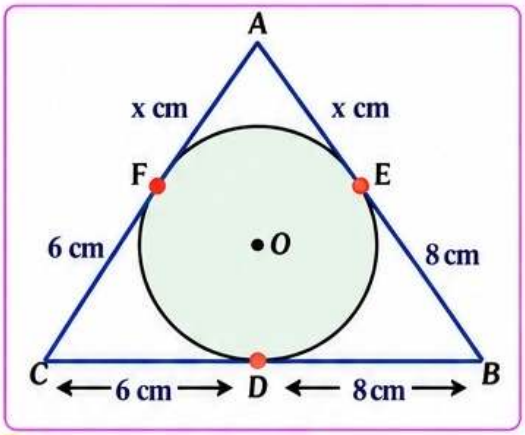
$= 2(8+6+14) = 2 \times 28 = 56 = 4(x+14)$ ...(ii)

From (i) and (ii), $\sqrt{48x(x+14)} = 4(x+14)$

$\Rightarrow 48x(x+14) = 16(x+14)^2$

$\Rightarrow 3x = x + 14 \Rightarrow x = 7$

Hence, AB = $x + 8 = 15$ cm and AC = $x + 6 = 13$ cm.



Concept Clarity

- Tangents from an external point to a circle are equal.
- Use given lengths to express sides in terms of a variable.
- Area of a triangle with an incircle can be found using Heron's formula.
- Also, Area of $\Delta ABC = 2r(a+b+c)$, where r is the inradius.
- Equate the two expressions for area to find the unknown.

Q13. **Prove that opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.**

Sol.

Let ABCD be a quadrilateral circumscribing a circle with centre O.

Let the circle touch AB, BC, CD and DA at P, Q, R and S respectively.

Join OA, OB, OC, OD and OP, OQ, OR, OS.

Since tangents from an external point are equal,

OP bisects $\angle AOB$, so $\angle 1 = \angle 2$.

OQ bisects $\angle BOC$, so $\angle 3 = \angle 4$.

OR bisects $\angle COD$, so $\angle 5 = \angle 6$.

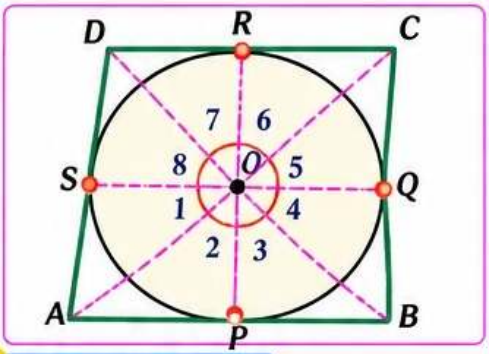
OS bisects $\angle AOD$, so $\angle 7 = \angle 8$.

Now, $\angle AOD + \angle BOC = (\angle 1 + \angle 8) + (\angle 4 + \angle 5)$

$= (\angle 1 + \angle 4 + \angle 5 + \angle 8) = \frac{1}{2} \times 360^\circ = 180^\circ$

Similarly, $\angle AOB + \angle COD = 180^\circ$.

Hence, opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.



Concept Clarity

- Radius to the point of contact is perpendicular to the tangent.
- The radius to a side bisects the angle between the radii to its adjacent sides.
- The sum of angles around a point is 360°.
- Therefore, opposite sides subtend supplementary angles at the centre.